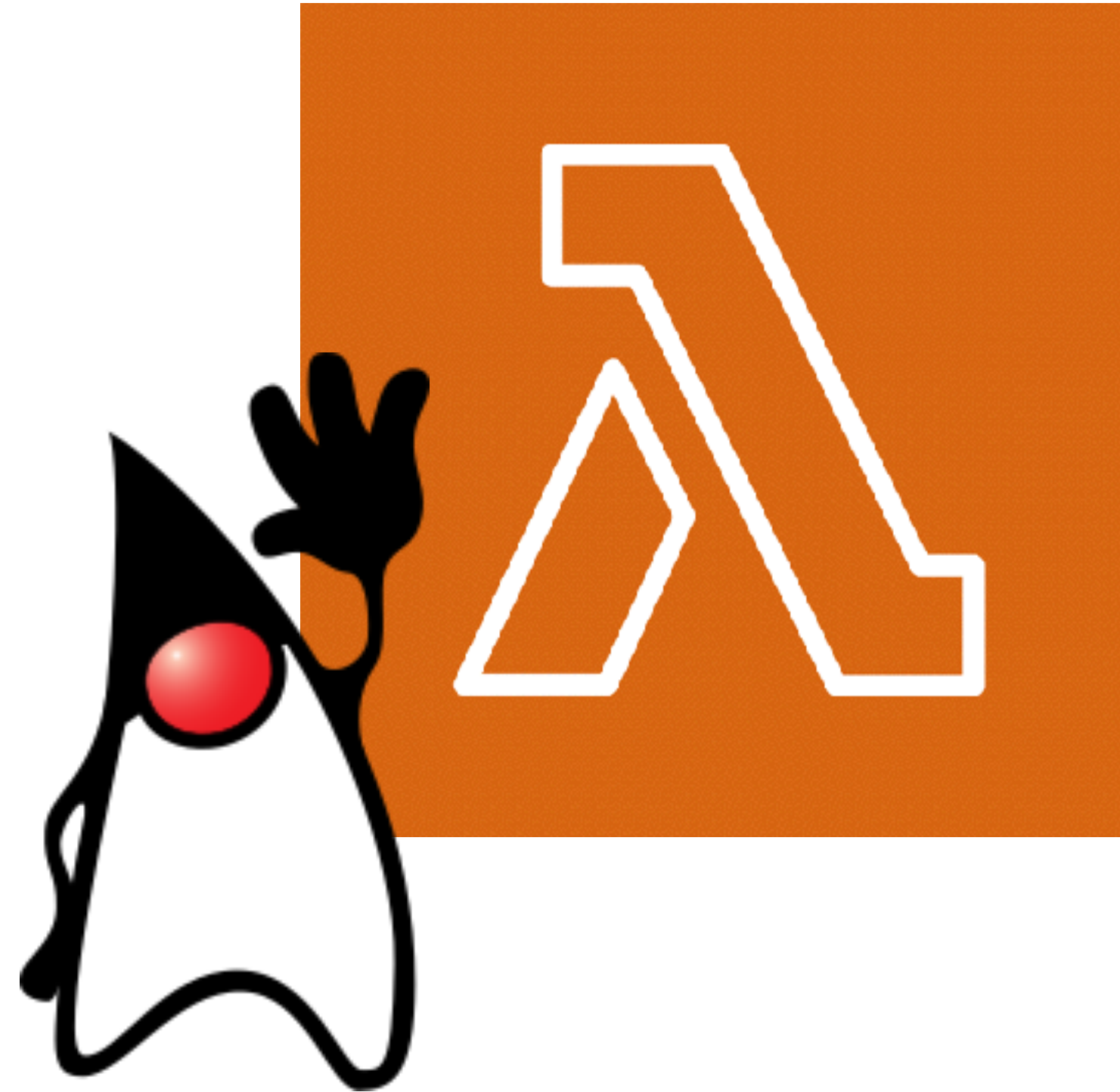




AWS Lambda SnapStart

Optimizing cold starts for Java functions

About me

- CapTech - Director | Architecture
- 19 years in the IT industry
- Architecture, cloud and development
- Family, video games, sports, books



What is *AWS* Lambda?



What is AWS Lambda?

- Compute service
- No provisioning or managing servers
- Provide your code or container image
- Automatic scaling
- Pay by the millisecond

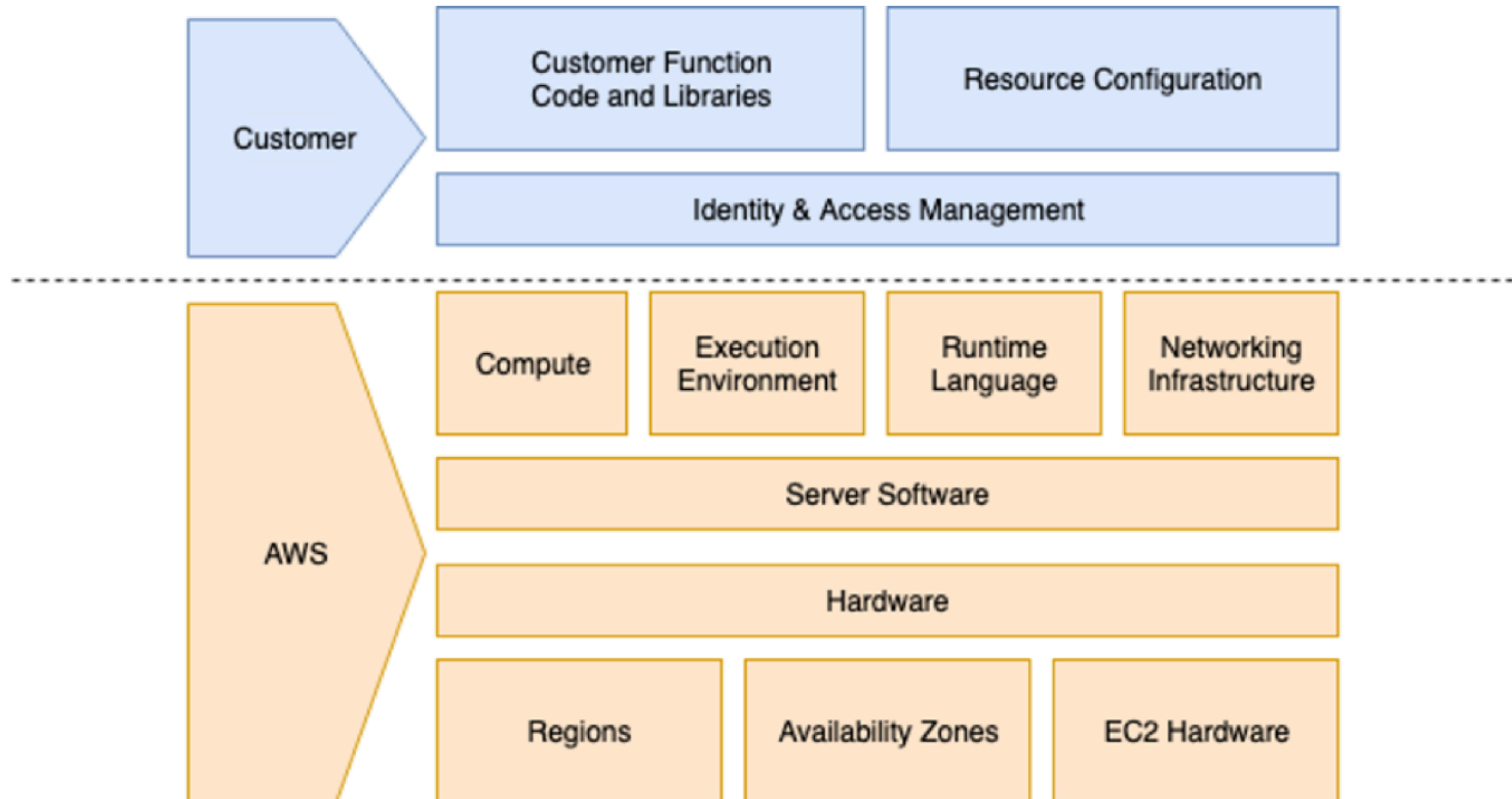


AWS Lambda In-Depth

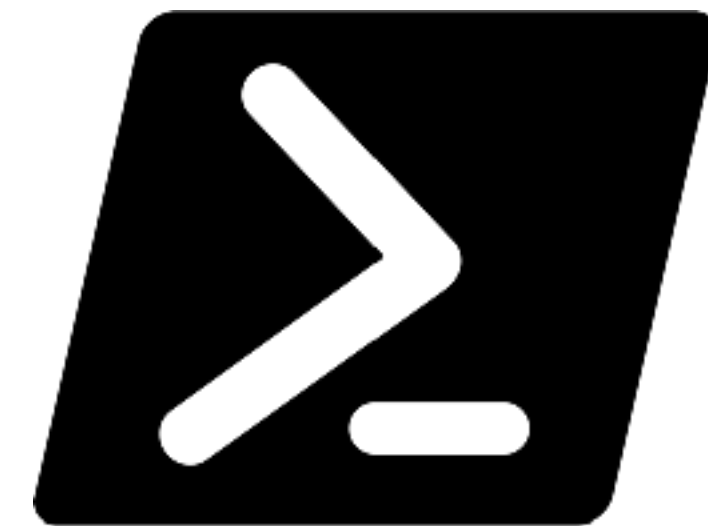
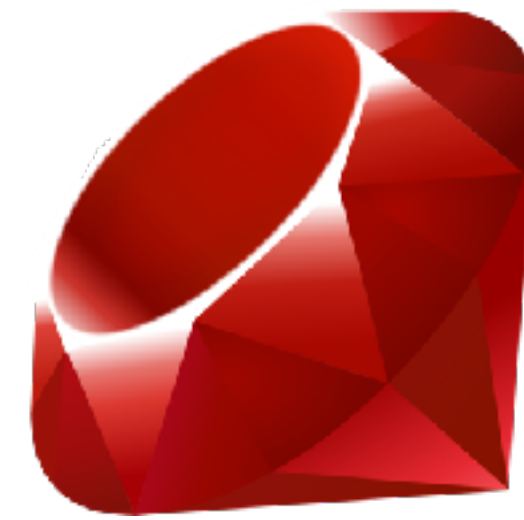
- Runs in an isolated execution environment
- You allocate memory
- AWS allocates CPU proportionally
- Temporary storage from 512Mb up to 10,240Mb
- Function timeout 900 seconds (15 minutes)
- IAM based permissions



AWS Shared Responsibility Model

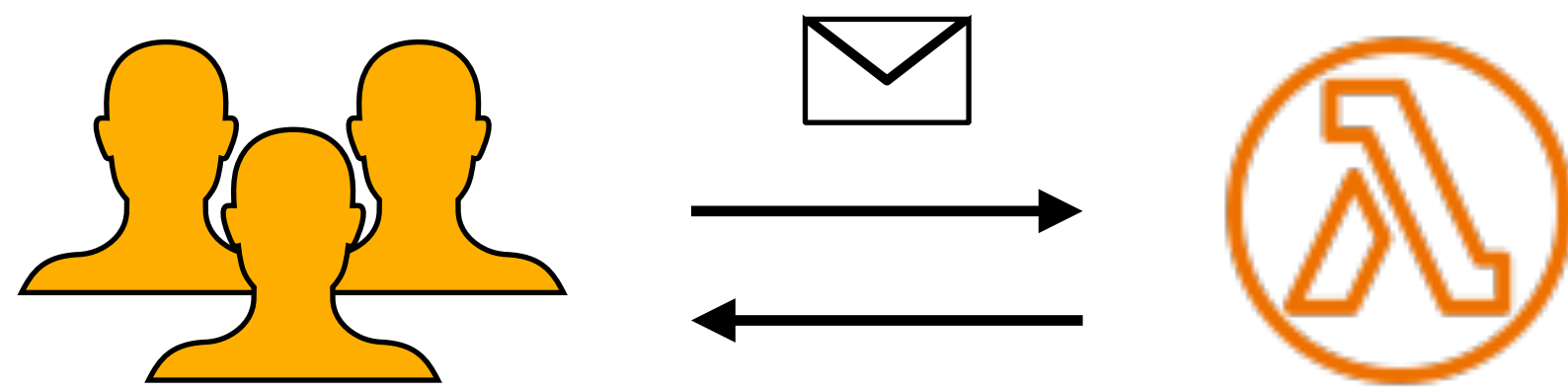


AWS Lambda Runtimes

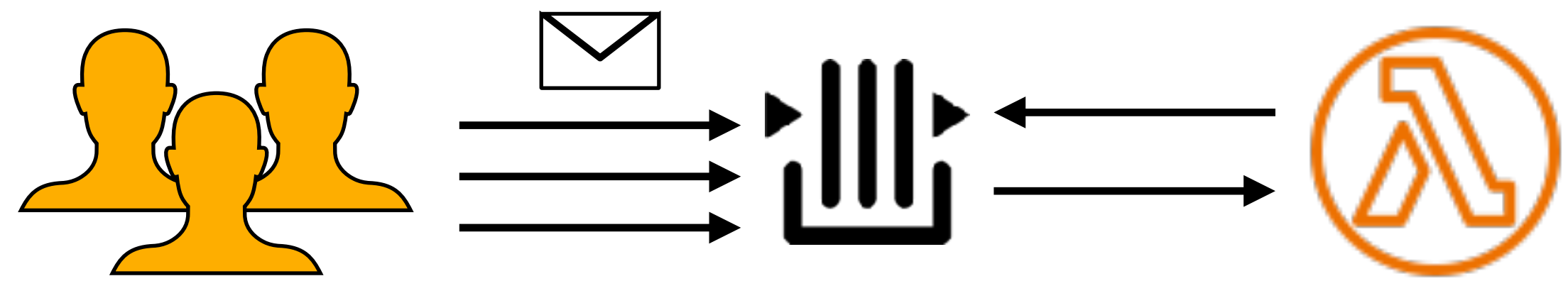


Invocation Models

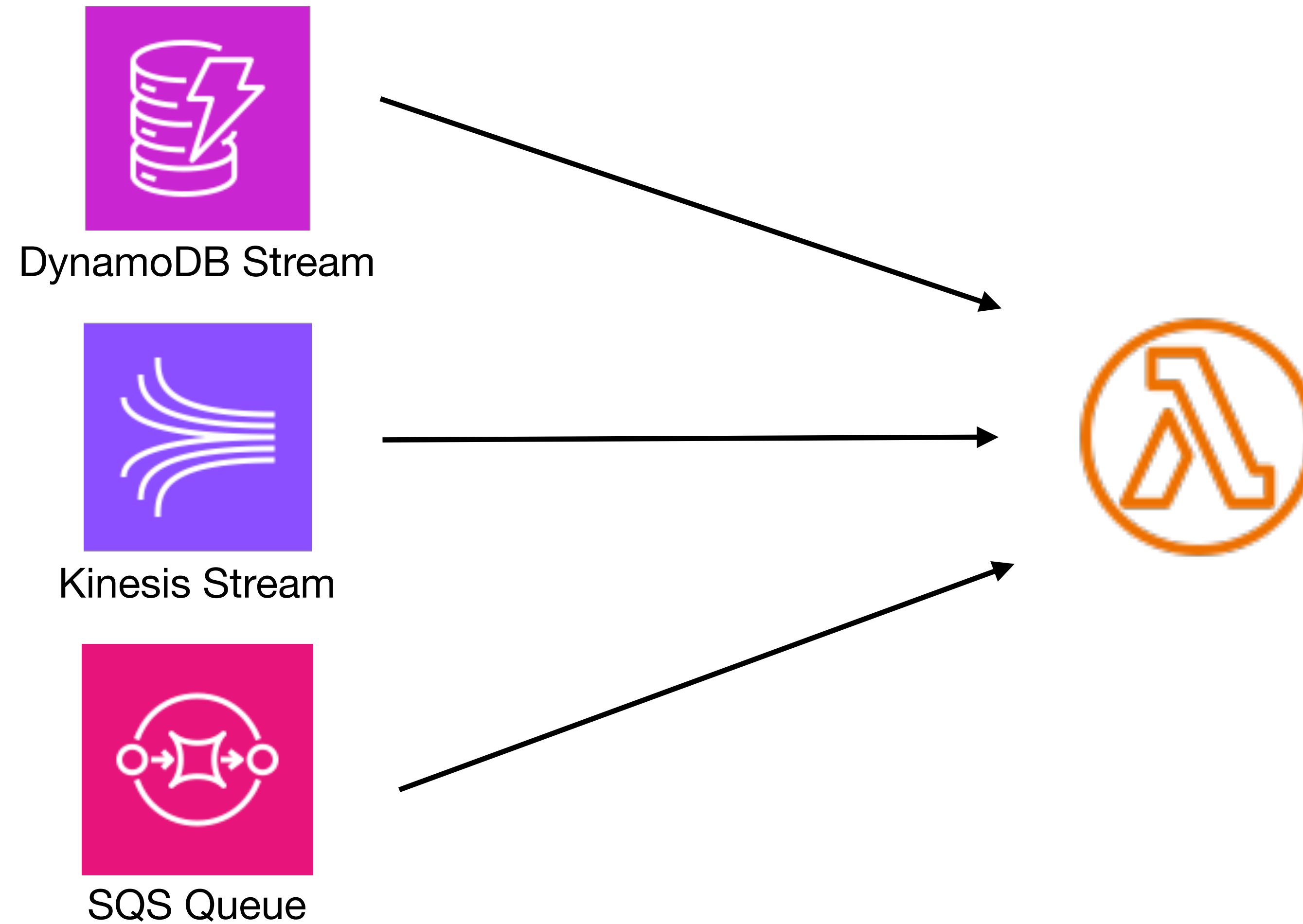
Synchronous



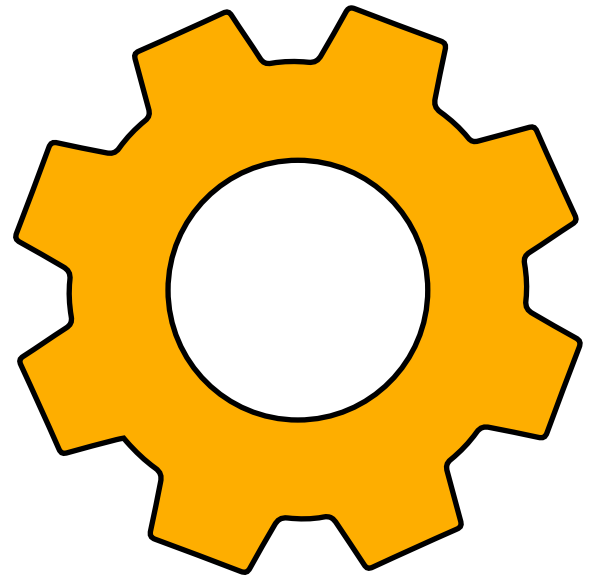
Asynchronous



Event Source Mapping



Use Cases



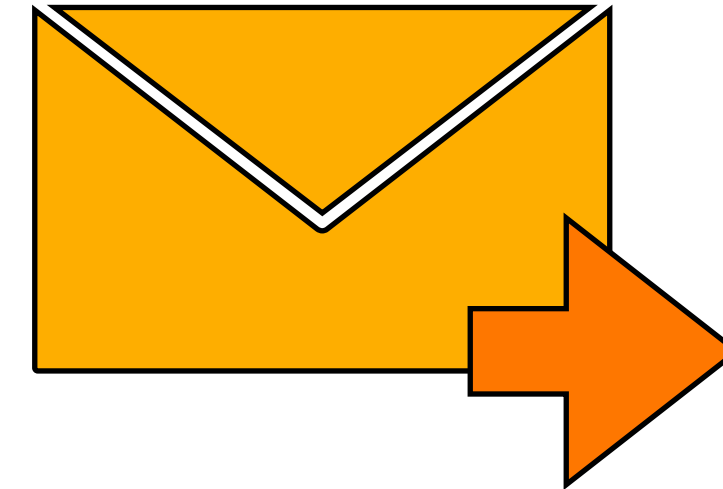
Automation



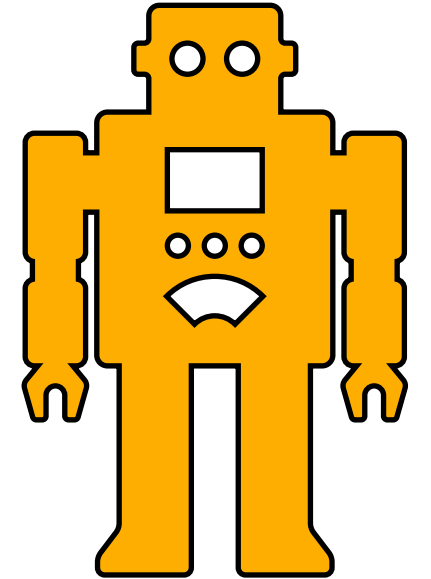
Data
Processing



Web
Applications

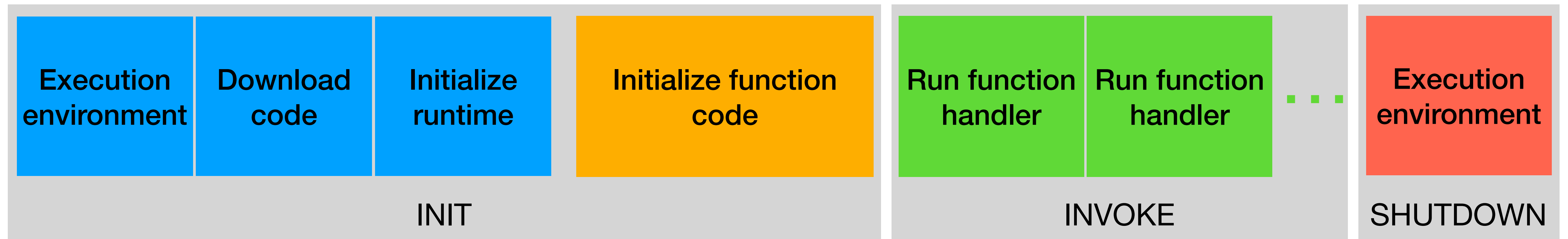


Event-Driven

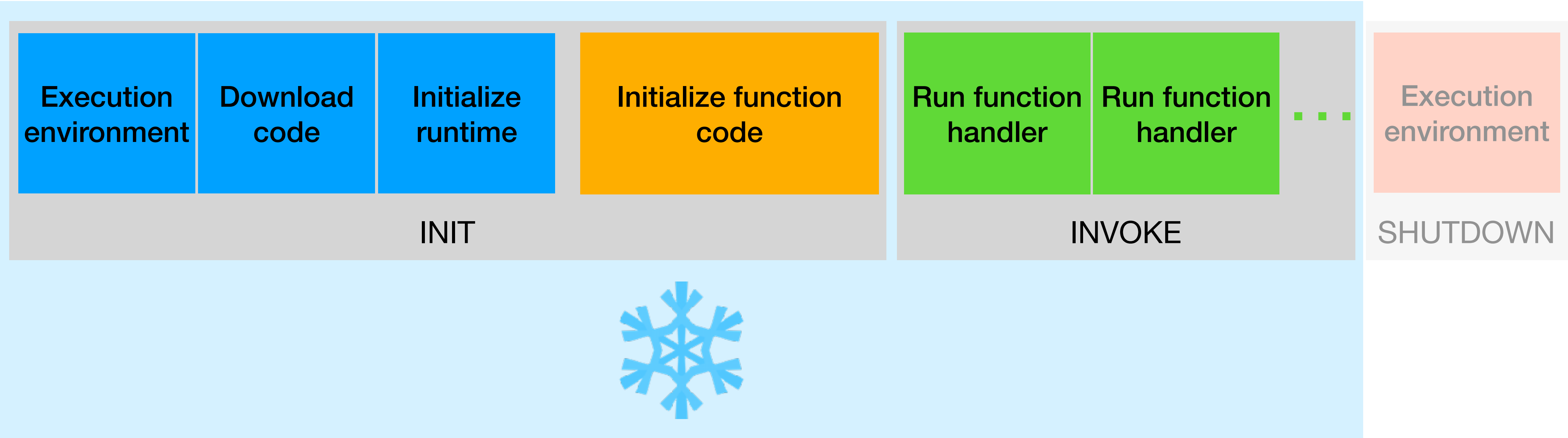


Machine
Learning

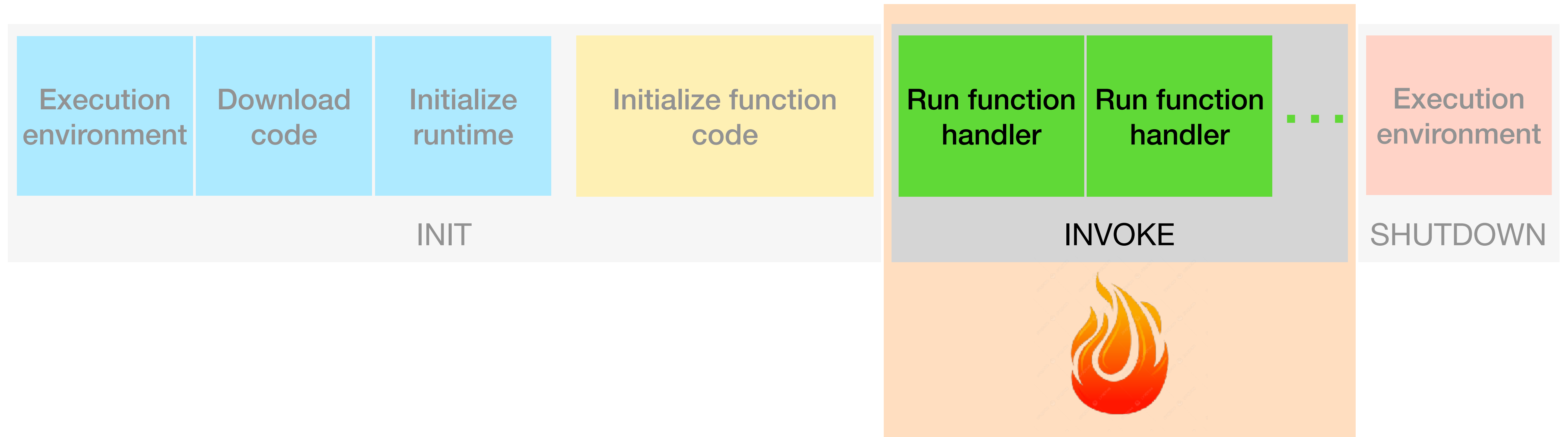
AWS Lambda Lifecycle



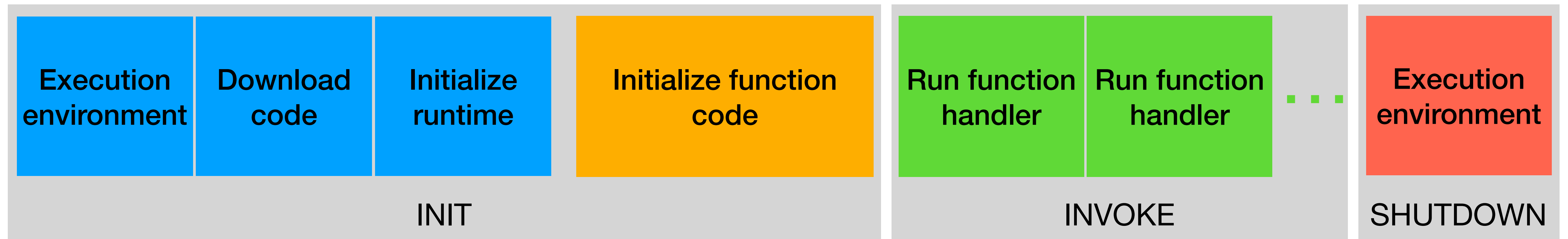
AWS Lambda Cold Start



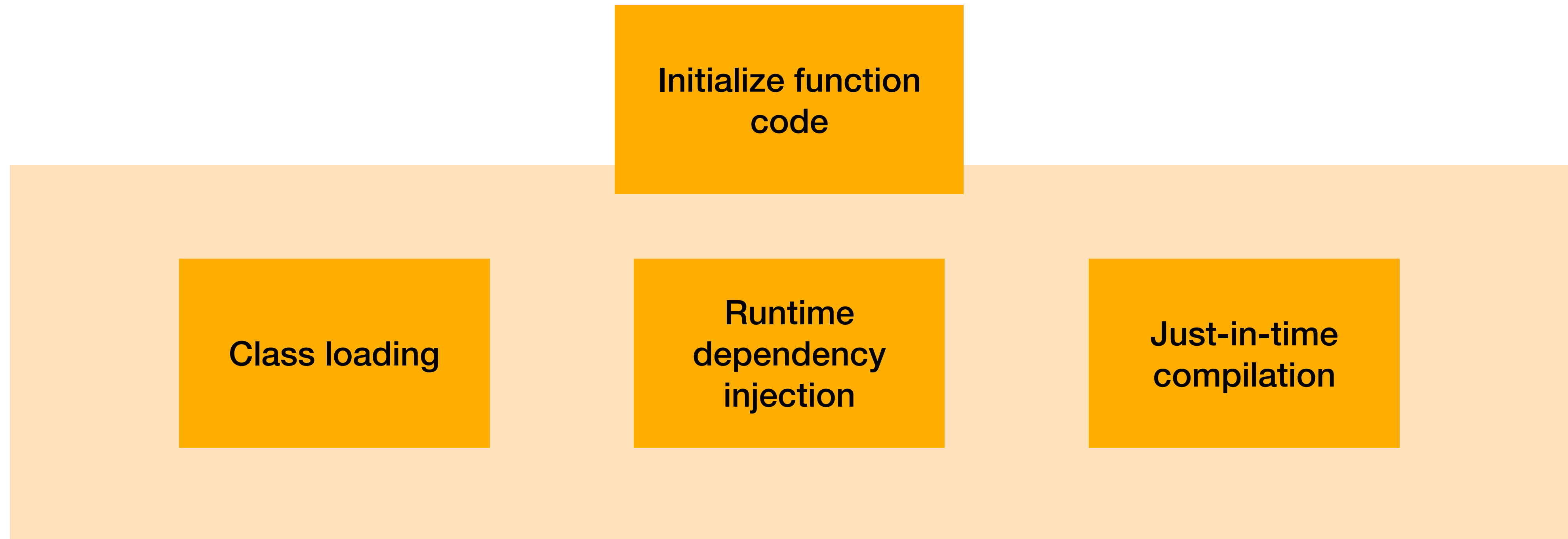
AWS Lambda Warm Start



AWS Lambda using Java



AWS Lambda using Java



Optimization Options

Provisioned Concurrency	—————→	High optimization, low complexity - Cost increase
Lightweight dependencies	—————→	Moderate optimization - Moderate complexity
Modern frameworks	—————→	Moderate-high optimization - Moderate-high complexity
GraalVM	—————→	High optimization - High complexity

AWS Lambda SnapStart

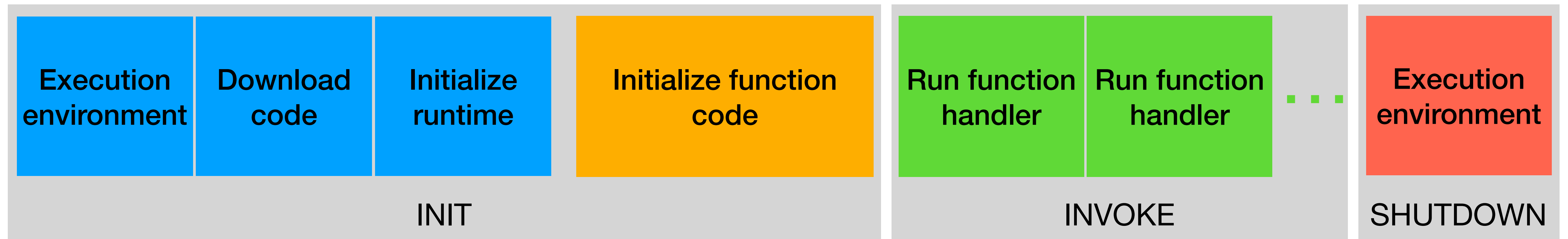


AWS Lambda SnapStart

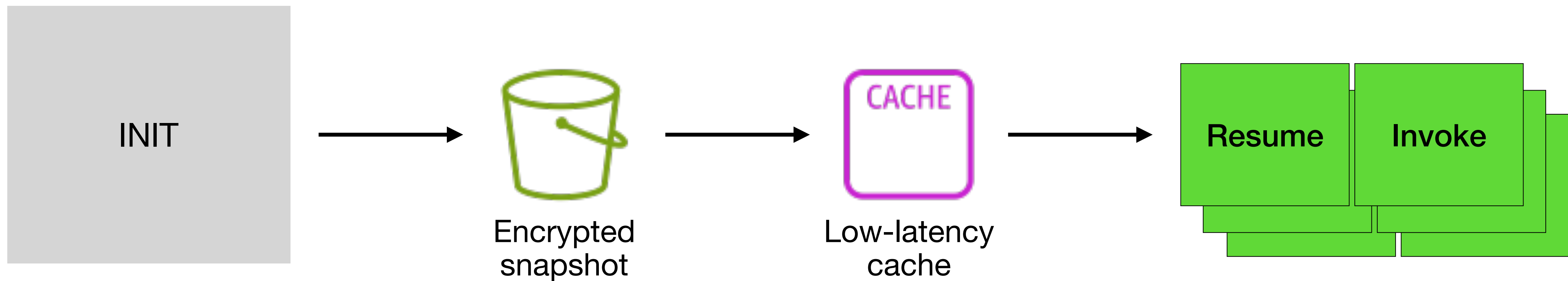
- Sub-second startup performance
- Typically no code changes needed
- Powered by Firecracker
- MicroVM snapshots technology



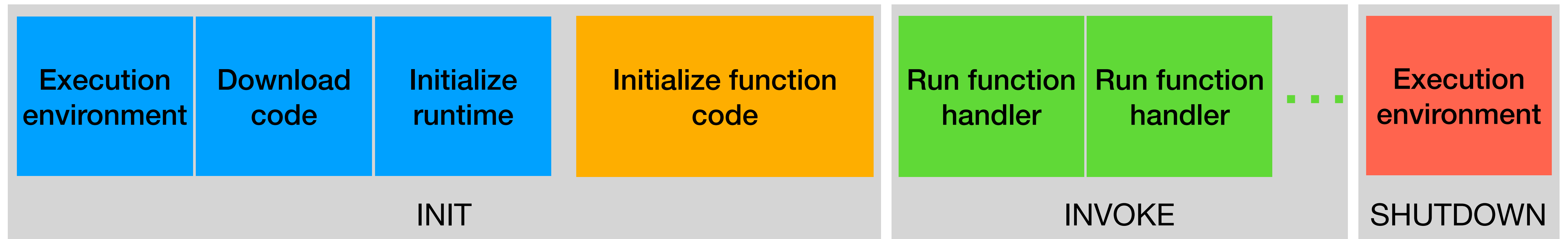
AWS Lambda SnapStart Overview



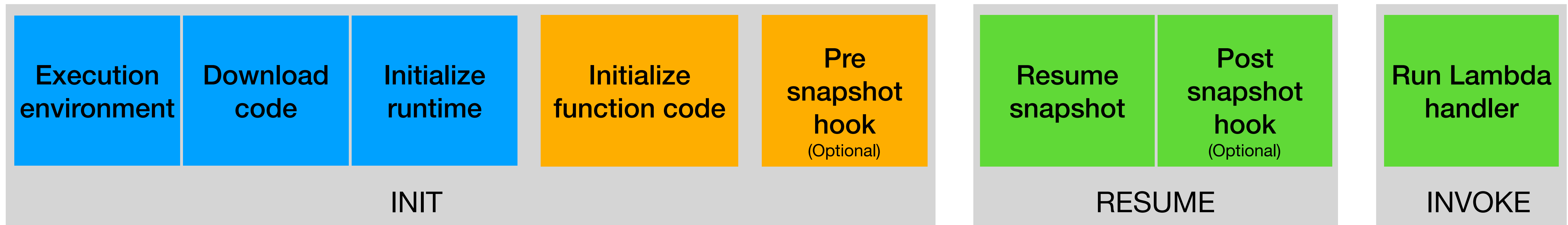
AWS Lambda SnapStart Overview



AWS Lambda SnapStart Overview



AWS Lambda SnapStart Overview



Considerations



Network Connections

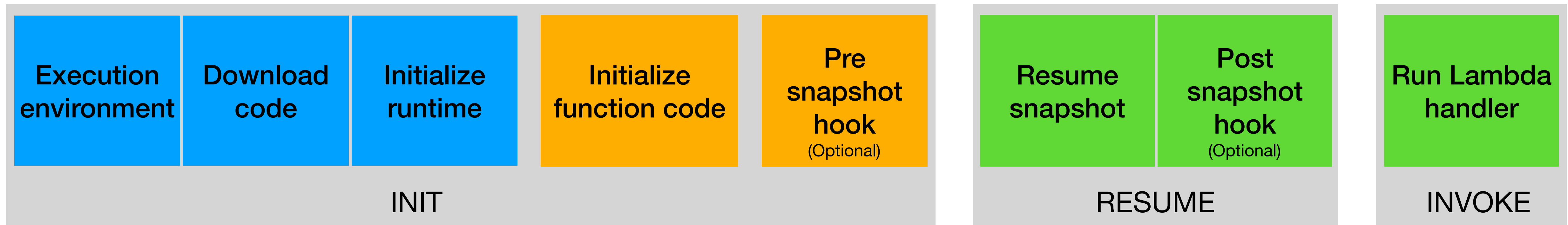


Ephemeral Data



Uniqueness

AWS Lambda SnapStart Hooks



AWS Lambda SnapStart Hooks

Pre
snapshot
hook
(Optional)



Cleanup - Initialization

Post
snapshot
hook
(Optional)

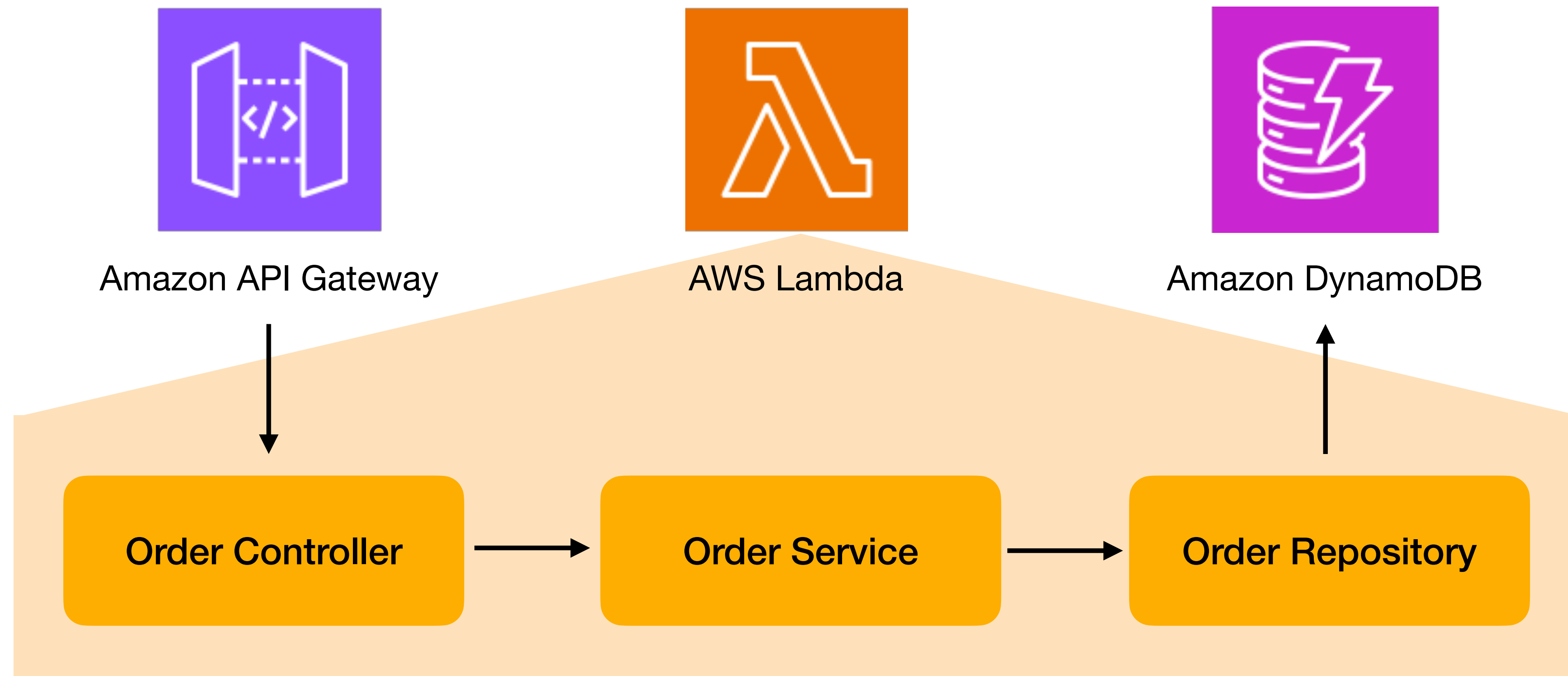


Restore network connections - Restore ephemeral data

Best Practices

- Re-establish network connections (handler/after-restore hook)
- Generate a unique execution environment ID (handler/after-restore hook)
- Avoid binding connections to fixed source ports
- Avoid using Java DNS cache

AWS Lambda SnapStart Demo



Repository and Slides

